

Akhil Yerrapally

Project Portfolio



B.S. Computer Engineering · University of Illinois Urbana-Champaign · Class of 2029

Seeking: Summer 2027 Internship – Embedded Systems, Controls, and Hardware

In this Portfolio

01: MPPT Solar Charge Controller

Power Electronics · PWM · DC-DC · Control · Telemetry

02: Self-Balancing Robot

ESP32 · C++ · PID Control · I²C · Sensor Fusion · Encoders

03: Intelligent Dog Door

Arduino · State Machines · C++ · SPI · Stepper Control

04: Additional Builds

Light-seeking Car · Resistor-key Circuit

Skills & Domain Expertise

Languages/Software: C, C++, Python, Git, Linux, KiCAD, Altium Designer, Fusion 360

Embedded: ESP32, Arduino, Raspberry Pi, CAN, I²C Devices, SPI, UART, PWM, ADC, hardware interrupts / ISRs, quadrature decoding, non-blocking state machines

Controls: PID tuning, complementary filtering, IMU sensor fusion, stepper motor control, cascade loop architecture, analog filtering, MPPT algorithms, closed-loop regulation

Hardware: Motor drivers, RFID, voltage/current sensing, DC-DC converters, soldering, breadboard prototyping, decoupling & signal integrity, multimeter, oscilloscope

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MPPT Solar Charge Controller

A custom buck-converter-based charge controller that dynamically tracks maximum power point using perturb-and-observe logic.

My Role: Solo Build

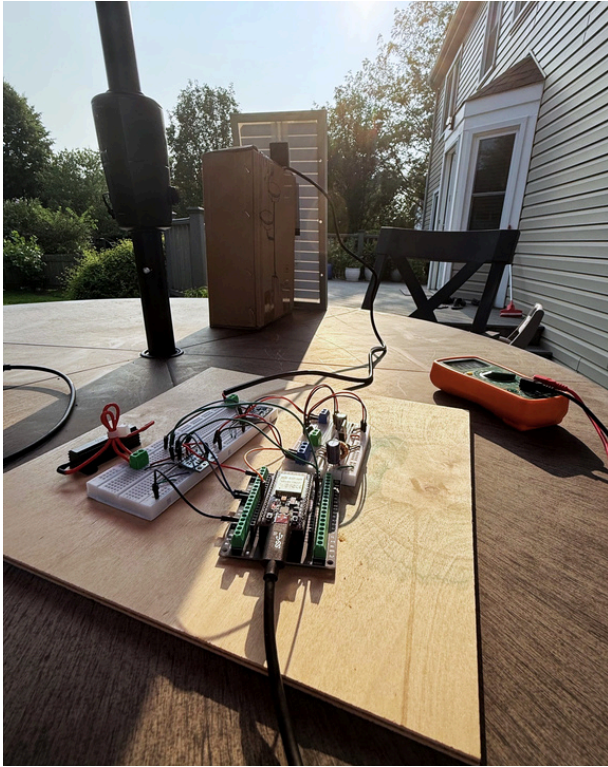


FIG. 1: Final prototype with MPPT charge control: closed-loop power tracking with an asynchronous, custom-built DC-DC converter

WHAT IT DOES

- Two INA226 monitors read voltage & current on the panel and load sides, feeding live measurements to an ESP32 over I²C
- A perturb-&-observe loop taps the PWM duty cycle: it watches and steps towards whichever direction raises power output
- A custom buck converter lowers the panel voltage while a level shifter allows the 3.3 V ESP32 to safely control the power switch

DESIGN & PROTOTYPE

- **Topology & impedance matching:** An asynchronous buck tracks $\text{load} \div D^2$, so sweeping duty sweeps the apparent load and gives the tracker a continuous peak-power output pointer
- **Gate drive:** An NPN transistor pulls the P-MOS gate below the panel rail while two series zeners clamp gate-to-source voltage within the ± 20 V limit as overload protection
- **Measurement & control:** Two INA226 monitors bracket the converter on independent I²C buses, so the panel-side reading drives the P&O loop while both together give live efficiency
- **Bring-up debugging:** Performed staged power-stage bring-up to isolate control vs. power faults; verified PWM switching and monitored gate-to-source voltage waveforms to prevent shoot-through before applying high supply voltage

RESULTS

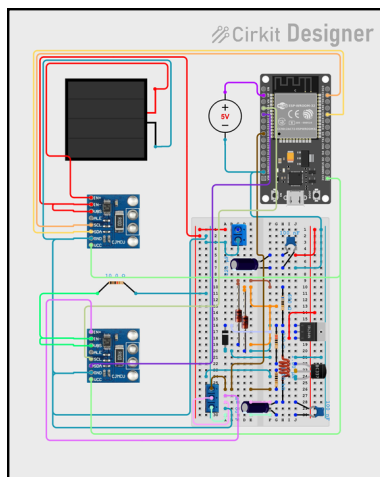


FIG. 2: Wiring diagram: Async. custom buck, sensors, & control

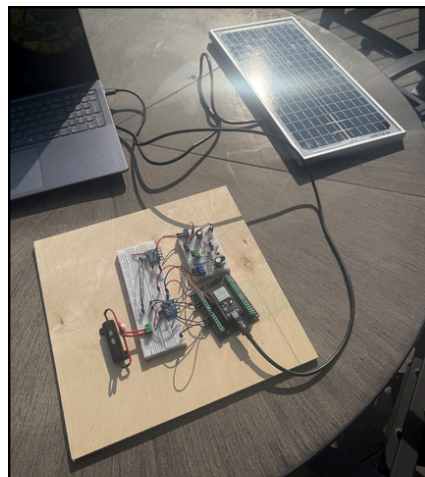


FIG. 3: As-built extracted circuit and measurement tools required

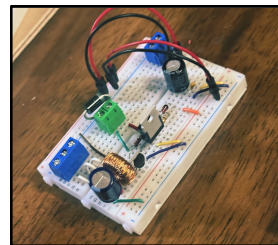
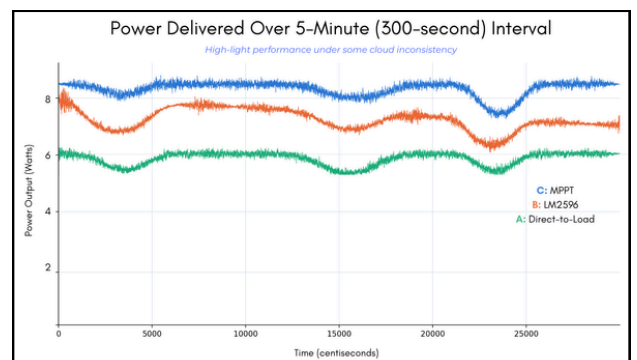


FIG. 4: Custom async. DC-DC buck converter

Testing achieved a **15% power delivery increase** over a commercial buck converter & **42% increase** over a direct-to-load (33 Ω Power Load) system in dynamic solar conditions



- Stepping duty across its range locates the true peak, then P&O continuously converges on that same point

Self-Balancing Robot

An inverted-pendulum robot that holds itself upright on hardwood, carpet, and bumpy terrain, and recovers from a physical shove.

My Role: Solo Build

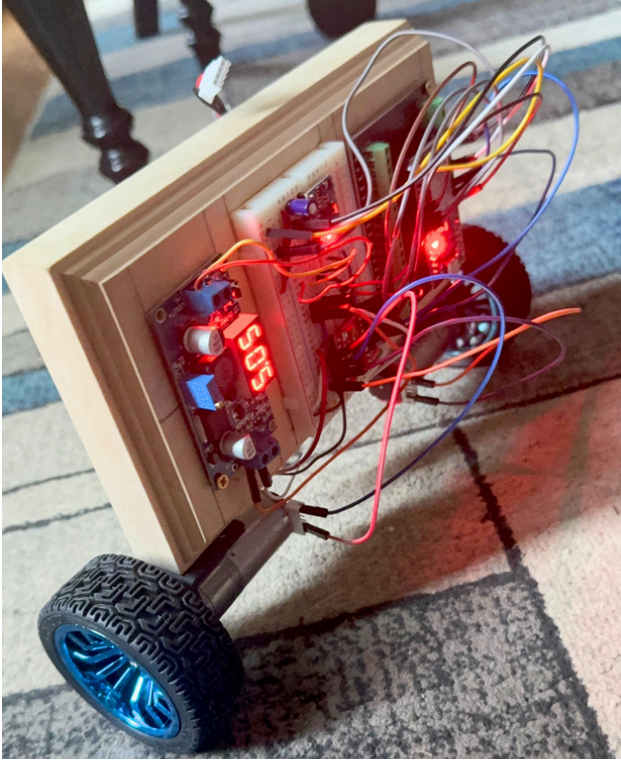


FIG. 1: Final prototype (P4) actively balancing: single PID loop for control logic, 3S LiPo being the central, rechargeable power source

WHAT IT DOES

- MPU-6050 IMU feeds a complementary filter that fuses gyro & accelerometer data into a pitch estimate
- A PID loop compares that pitch against a calibrated upright setpoint and computes a correction
- A TB6612FNG drives two encoder gear motors to hold the robot upright and recover from disturbances

DESIGN & PROTOTYPE

- **Design pivot:** Four chassis iterations – the fix was raising the center of mass. Prototypes 2 and 3 sat too low to absorb terrain without oscillating out of control
- **Control law:** Single PID loop on pitch error with an added cubic term, so correction scales up as tilt grows instead of responding uniformly. Encoder RPM decoded in hardware ISRs at 4x quadrature resolution
- **Angle estimation:** Complementary filter at $\alpha = 0.98$, IMU registers read directly over I²C. Mounted centered over the axle so movement doesn't corrupt the estimate
- **Power & bring-up:** 3S LiPo to the driver, buck-converted to 5V for the ESP32, common ground. Motor B originally on GPIO12 (strapping pin), broke flashing until remapped

RESULTS

Balancing: 100% success (3 settled indefinitely, with the remaining 2 tests lasting over 100 seconds)

Terrain: Final prototype correctly balanced on flat hardwood & uneven carpet

Recovery: 3/5 tap trials recovered within 5 seconds of the disturbance; other 2 trials oscillated & gave out

Bottleneck: Motor torque and overall chassis rigidity

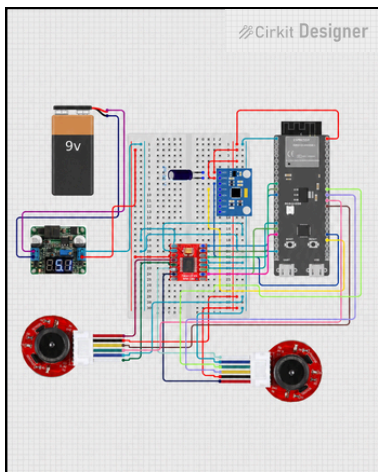


FIG. 2: Wiring diagram: Logic, sensors, & power distribution

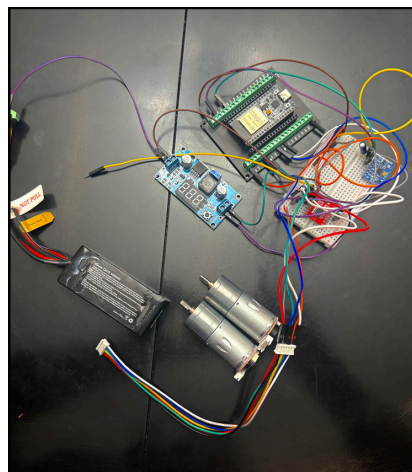


FIG. 3: As-built circuit (encoders & temp. 2S LiPo) without chassis

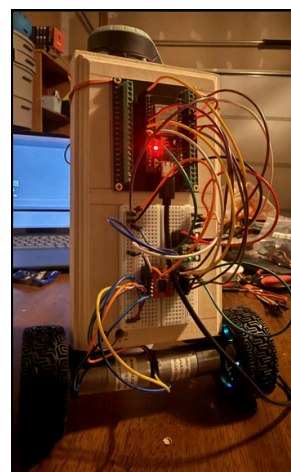


FIG. 4: Early prototype; failed recovery tests

- A cascade variant (double-loop) is also running on the same hardware: an outer angle loop setting a target wheel speed, with an independent inner RPM loop per motor

Intelligent Dog Door

An RFID-authenticated pet entry that opens only for a tagged animal and mechanically locks itself shut against anything else.

My Role: Software and Electronics Lead



FIG. 1: Installed prototype: worm drive at left, door in the open position with both sensors activated

FIG. 2: Installed prototype: Idle position, door closed

WHAT IT DOES

- A PIR motion sensor wakes the system when an animal approaches the door
- An RC522 reader authenticates the pet's collar tag over SPI; unknown tags are ignored
- A NEMA 17 stepper swings the door open, then closes it into a self-locking gear state

DESIGN & PROTOTYPE

- **Design pivot:** BLE/actuators replaced with RFID & worm gear – offering higher ID resolution, automatic locking
- **Firmware:** Non-blocking C++ state machine – *AccelStepper* commands the A4988 in open-loop step counts so the main loop keeps polling sensors instead of stalling inside *delay()*
- **Power & signal integrity:** 5V logic interfaced to a 3.3V SPI peripheral, with the motor on a separate 9V rail and a 100 μ F bulk cap to keep driver switching transients off the logic supply. Set coil current by tuning the A4988 V_{ref}
- **Mechanism:** A worm-to-spur gear train is non-backdrivable, so the closed door resists being forced open with the motor unpowered

RESULTS

92% final product reliability score after administering various functionality tests

Final prototype withstands harsher weather conditions

Live Data: The program accurately logs entry and exit times on the Arduino's local web server

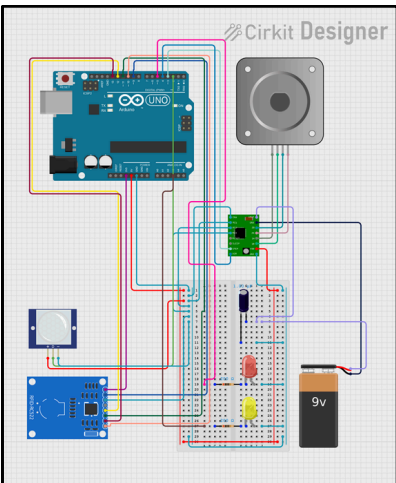


FIG. 3: Wiring diagram: Logic, driver, and 9V motor power rail

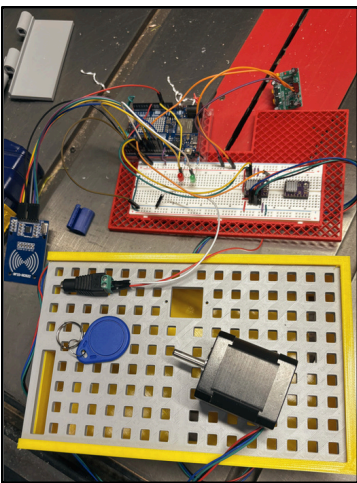


FIG. 4: As-built central circuit with vented enclosure panel

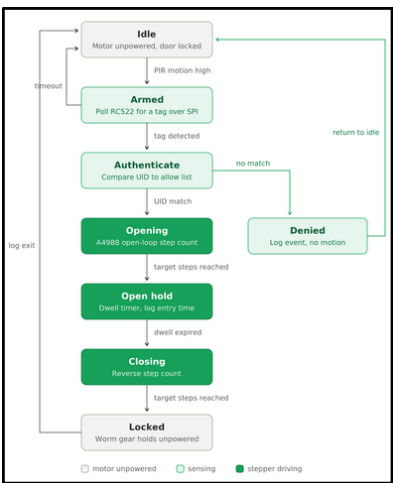


FIG. 5: Dog door prototype's state machine flowchart

- Built by a 4-person team over 6 months for Project Lead the Way (PLTW) capstone project.
- Final prototype ran reliably through demo-day testing on passive cooling via the vented enclosure.

Additional Builds

Wheatstone Bridge • Voltage Dividers • Comparators • MOSFET Switching • Discrete Logic • RC Timing • Photoresistive Sensing • Oscilloscope Debugging

RESISTOR-KEY LOGIC CIRCUIT

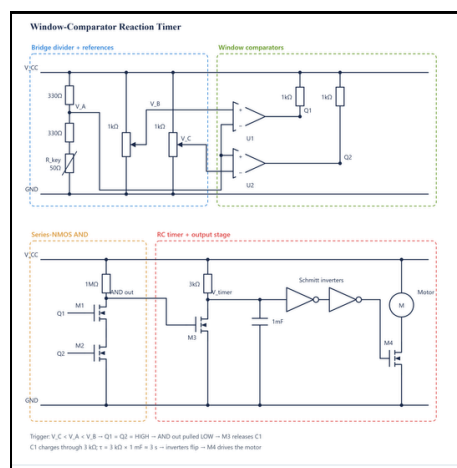


FIG. 3: Schematic: Wheatstone bridge, dual comparator, NMOS AND, RC Timer

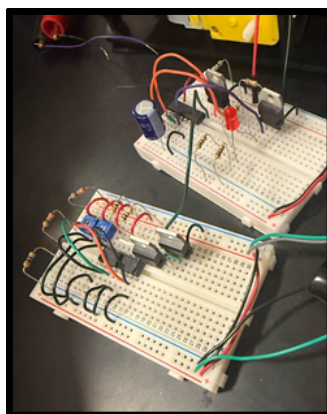


FIG. 4: As-built resistor-key system: Breadboard setup with potentiometer tuning and timed MOSFET motor gate drive

DESIGN & PROTOTYPE

- **Detection window:** A Wheatstone bridge puts V_A at 4.81 V with the correct 50 Ω key; two pots set comparator references 60 mV apart. Wrong keys miss it, with 0 Ω dropping V_A to 4.32 V and 1 k Ω pushing it to 7.13 V
- **Discrete logic:** The AND stage is two series N-channel MOSFETs rather than a logic IC, measuring 8.9 V HIGH and 0.1 V LOW
- **Anti-brute-force timing:** A 3 k Ω / 1000 μF RC ($\tau = 3\text{ s}$) gates the motor, with a shunt nMOS dumping the cap the instant the AND output drops. Sweeping a pot never accumulates charge, so the key must hold correct for 3 s

LIGHT-SEEKING VEHICLE

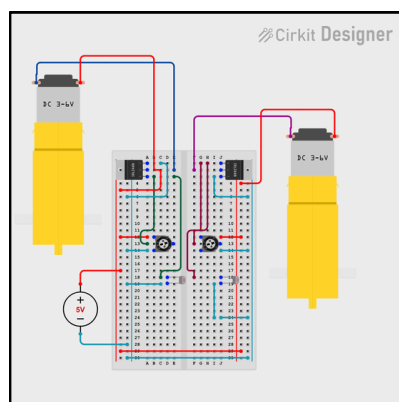


FIG. 1: Wiring diagram: Breadboard layout and dual-channel nMOSFET speed control circuit for light-seeking behavior

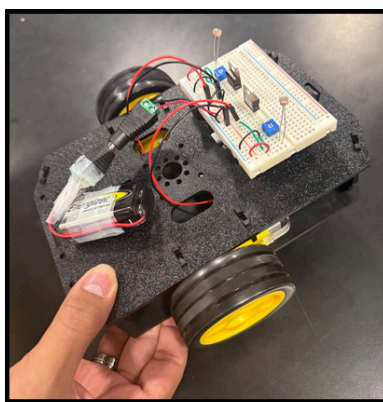


FIG. 2: As-built prototype: Breadboard circuit, sensor roles, dual DC motor assembly, working under high-light

DESIGN & PROTOTYPE

- **Analog speed control:** A photoresistor / potentiometer divider drives each nMOSFET gate directly, so motor speed scales continuously with the environment's brightness rather than simply switching at a threshold
- **Channel matching:** Sensitivity mismatch between the two channels translated into steering bias, so the early build just spun in place. Pots required tuning to confirm both gates tracked on the scope
- **Discrete logic:** Turning, stopping, and variable speed all emerge from component placement. The control logic is based on the resistor that sits on top of the divider